Modality Fairness via OGR-Driven Gradient Harmonization: Dynamic Loss Balancing Against Gradient Starvation

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Abstract

Multi modal learning systems are likely to be stuck in one of such traps and it is known as the Modality Laziness trap whereby they over-respond to the best signal in most instances which happens to be the best signal. Visual signal and ignore the weak signals such as the Audio or Text signal. The unified models are their possible limited by this optimization bias. Although fixes such as Gradient Blending or OGM-GE currently exist with most of them being based on expensive pre-computations or noisy and unstable noise. Insertion which is incapable of following the dynamics of training. In this paper, the Modality is presented. Fairness Controller (MFC), a simple and easy to use solution. The idea behind the MFC is to look at the Overfitting-to-Generalization Ratio (OGR) of each modality and dynamically vary the level to which each modality affects the learning process. We were able to test on a controlled synthetic benchmark, and a real-world CREMA-D emotion recognition (7,442 mel spectrogram samples, 6 classes). MFC-A (our best variant) reduces modality imbalance by 38.4% (OGR gap 7.89→4.86) at the synthetic benchmark and enhances robustness by +2.3% at σ=1.0 noise, with zero additional FLOPs over baseline training. At actual CREMA-D data the MFC is as well as accurate as at the baseline and diminishes the OGR dispersion by 38.9 percent and beats OGM-GE by +3.1% accuracy- without incurring any overhead of noise injection.

Index Terms— Multi-modal Learning, Modality Laziness, Gradient Starvation, Loss Balancing, Overfitting-to-Generalization Ratio (OGR), Emotion Recognition, Dynamic Modulation, Modality Fairness Controller (MFC).

1 Introduction

The world is tasted in the colourful mixture of impressions. This is what ideally our AI models should do. Nevertheless, such multimodal architectures are usually difficult to train, and powerful ones like Unified Transformers [1], Image Bind [2], CLIP [3] and Flamingo [4] these become lazy. When one of the modalities (say an image) is easier to learn than the other (say audio) then the model neglects the more difficult task. This will be called a Modality Laziness or Gradient Starvation [5]. The visual features too are generally means puny and arranged; this presents a simple access point leading to the optimizer. Noisey audio or text may be less or excessive. This means that the high visual gradients would tend to absorb the common weights in the model at almost full costs and the other modalities are under-developed. In the real world such as an autonomous car that may have to hear a siren that it cannot see this blindness can be hazardous. Although the various fixes have been enumerated in various recent surveys, like the Balance Benchmark [6], most of them are burdensome. Others must be taught a high number of models to get appropriate weights or simply strive to resolve the problem by introducing random noise, again a common hit-or-miss method, like OGM-GE [7]. We propose a more idealistic one the Modality Fairness Controller (MFC). Instead of simply slowing down the fast learner, we pose a simple question: Is this modality in fact generalizing or is it simply memorizing? We can take the first step to having each modality pull its own weight with respect to the actual unified representation by taking the cue of limiting our learning rates by the Generalization Gap.

The visual features are also generally small and arranged and this provides a simple pathway to the optimizer [5]. Well-conditioned loss surface Vision features dense, spatially coherent, low-noise (e.g. CIFAR-10 [8]) features. Audio characteristics, such as speaker-varying, temporally noisy (e.g., CREMA-D [9]) create a more rugged and high-variance landscape. Thus, the powerful visual gradients are inclined to attract the shared weights in the model almost entirely and the remaining modalities are under-developed. The implications cannot be seen in normal assessment but are disastrous when stressed modality. The baseline performance of 83.5% accuracy

falls to 48.2% accuracy with the addition of Gaussian noise (at 1.0) to visual input (a 35-point drop) because the audio encoder was never developed. In autonomous driving, fog or glare produces the same failure. In medical imaging, corrupted scans leave the model unable to fall back on clinical text it never learned.

Root Cause

The visual features are also normally compact and organized, which offers an easy route to the optimizer [5]. Vision features—dense, spatially coherent, low-noise (e.g., CIFAR-10 [8])—produce a well-conditioned loss surface. Audio features—speaker-dependent, temporally noisy (e.g., CREMA-D [9])—produce a rougher, high-variance landscape. Consequently, the strong visual gradients tend to pull the shared weights in the model nearly completely and the rest of the modalities become underdeveloped.

The consequences are invisible during standard evaluation but severe under modality stress. Our experiments show the baseline collapses from 83.5% to 48.2% accuracy when Gaussian noise ($\sigma = 1.0$) is added to visual input—a 35-point drop—because the audio encoder was never developed. In the real world such as an autonomous car that may have to hear a siren that it cannot see this blindness can be hazardous.

Limits of Existing Approaches

While Although recent surveys such as Balance Benchmark [6] have listed different fixes, most of them are cumbersome. Gradient Blending [10] trains M+1 separate models to compute static blending weights—mathematically principled, but M+2× the compute budget and unable to adapt as training dynamics shift. OGM-GE [7] injects Gaussian noise into the over-performing modality's backward pass—adaptive, but the noise variance σ must be hand-tuned and, as our CREMA-D results confirm, over-correction costs −3.1% accuracy. UWGA [11] weights by uncertainty—but overfit modalities are precisely the high-confidence ones, so it systematically boosts the wrong modality. All these methods share a deeper flaw: they monitor how fast each modality is changing, not whether that change generalizes. A large gradient norm can indicate productive learning or rapid memorization—gradient norms cannot distinguish the two.

Our Approach

We propose the Modality Fairness Controller (MFC), built on a novel per-modality metric: the Overfitting-to-Generalization Ratio (OGR).

$$OGR_m(t) = \frac{\mathcal{L}_m^{val}(t)}{\mathcal{L}_m^{train}(t) + \epsilon}$$

When $OGR \approx 1$, a modality is generalizing. When $OGR \gg 1$, it is overfitting. Unlike raw generalization gaps, OGR is scale-invariant and directly comparable across modalities with very different loss magnitudes. The MFC monitors per-modality OGRs via an exponential moving average and adjusts each modality's loss weight λ_m: throttle when $OGR > \tau_{hi}$, amplify when underfitting. No auxiliary networks, noise injection, or offline pre-computation are needed—overhead is $O(M \cdot H)$ scalars per epoch.

Contributions

OGR metric: a scale-invariant, gradient-free per-modality generalization diagnostic computed from standard train/val logs.

MFC: a closed-loop dynamic loss re-weighting controller operating in a single training run at negligible cost.

Validation on synthetic CIFAR-10/CREMA-D benchmarks and real CREMA-D emotion recognition: 38.4% OGR gap reduction, +2.3% robustness at σ=1.0, best Rob@0.5 (72.9%) across all methods—at zero additional FLOPs and 0.10% wall-clock overhead.

Scaling analysis: MFC gains grow with model capacity (+0.7% at 128-dim to +3.1% at 1024-dim), making it most valuable in the large-model regime.

2 Related Work

2.1 Static Re-weighting Strategies

Multimodal learning has been documented on the imbalance problem. The first to propose the calculation of optimal weights off-line was Gradient Blending [10]. They construct the weighting scheme in which unimodal and multimodal baselines are trained to converge to minimize the overfitting gap. Nevertheless, it is computationally prohibitive with large models, and scales in direct proportion to the number of combinations of modality. Other methods that are static such as UWGA (Uncertainty-Weighted Gradient Aggregation) [11] are trying to make use of task uncertainty, although they tend to fail when a single modality is inherently less noisy yet less informative (such as a black screen with perfect silence). Methods inspired by Focal Loss [12] have also been implemented but they do not deal with modality imbalance, but deal with class imbalance.

2.2 Dynamic Modulation Techniques

Dynamic methods seek to change weights dynamically. OGM-GE [7] keeps a check on the size of gradient updates. When the gradient norm of one modality is changing too rapidly (which means that the modality is converging too fast), OGM-GE adds Gaussian noise to the backward pass of that modality. Although effective, this is based on auxiliary noise generators that may destabilize

training and has hyperparameter sensitivity with respect to the noise variance.

GradNorm [13] was an early gradient-based approach that normalizes per-task gradient magnitudes to a shared target, preventing any single task from monopolizing the optimizer. **PCGrad** [14] detects conflicting gradients across tasks and projects them onto each other's normal planes to reduce interference. **AGM** [15] extends this idea to multimodal settings with per-modality adaptive gradient scaling calibrated against a reference modality. **Multi-loss gradient modulation** [16] jointly modulates multiple loss terms to keep modality-specific gradient magnitudes in balance throughout training. **OLM** [17] proposes online logit modulation to equalize modal contributions at the prediction level rather than the gradient level. **MMPareto** [18] frames multimodal optimization as a Pareto problem, using unimodal assistance signals to prevent the joint loss surface from collapsing onto the dominant modality. **AMSA** [19] demonstrates adaptive loss modulation in the sentiment analysis domain. PMR (Progressive Modality Reinforcement) [6], employs a momentum-based tracker to rampage the learning rate of the dominant modality. The alignment is modeled in MBPO [20] as a preference maximization problem which employs the reinforcement learning to punish hallucinations which are modality specific. We are different in the sense that we offer a closed-loop, parameter-free controller that is expressly on the basis of the Generalization Gap, which does not require the complexity of RL or is unstable in noise injection.

Method	Type	Dynamic?	Overhead
Gradient Blend [10]	Offline	No	High ($> 2\times$)
OGM-GE [7]	Optimization	Yes	Low ($< 5\%$)
PMR [6]	Data Aug.	No	Medium
MFC (Ours)	Optimization	Yes	**Negligible**

Table 1: **Comparison of Multimodal Imbalance Mitigation Strategies.** Our method combines the dynamic adaptability of OGM-GE with the low computational footprint required for modern large-scale pretraining.

2.3 Multimodal Fusion Architectures

The challenge of modality imbalance is intertwined with how modalities are fused. Early work such as **L3-Net** [21] demonstrated that audio-visual correspondence can be learned self-supervisedly from video, laying the foundation for studying cross-modal interaction. **Attention Bottlenecks** [22] introduced bottleneck tokens that force cross-modal information exchange at each transformer layer, reducing the risk of one stream being ignored. Datasets such as **VGGSound** [23] provide large-scale audio-visual benchmarks that expose imbalance under real-world noise conditions. Recent work on emotion recognition [24] employs recursive joint cross-modal attention, showing that carefully designed fusion mechanisms can mitigate unimodal dominance even without explicit loss reweighting. These architectural insights complement our OGR-based loss-level intervention: together, better fusion design and adaptive loss control form a complete defense against gradient starvation.

3 Preliminaries

3.1 Problem Formulation

Given a multimodal learning task, whereby there are modalities $\mathcal{M} = \{1, \ldots, M\}$ there are. Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)^N\}$ where $\mathbf{x}_i = \{x^m\}_{m \in \mathcal{M}}$ and x^m is a feature of the input (i.e. each image or audio waveform) on a specific modality (i.e. image or audio) and y_i is the target label. The objective is to train a joint representation function $f_\theta(\mathbf{x})$ with parameters θ that has the lowest expected risk:

$$\theta^* = \arg\min_\theta \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}}[\mathcal{L}(f_\theta(\mathbf{x}), y)] \tag{1}$$

In a late-fusion architecture, separate encoders E_m process each modality to produce latent embeddings $z_m = E_m(x^m)$, which are then concatenated or fused via a mechanism $\Phi(\{z_m\})$.

3.2 The Imbalance Challenge

The convergence rates of various encoders E_m are found to vary greatly empirically. Assuming $\nabla_{\theta_v}\mathcal{L} \gg \nabla_{\theta_a}\mathcal{L}$ then the visual encoder weights θ_v will drift far quicker to a local minimum than the audio encoder weights θ_a. After the visual encoder has reduced the loss to a sufficiently low level, the audio encoder is left in an undertrained condition. This is the nature of Modality Laziness.

4 Methodology: Modality Fairness Controller

4.1 The Gradient Starvation Phenomenon

The formalization of Gradient Starvation in multimodal settings was done by Pefrani et al. [5]. Suppose that there is a joint loss $\mathcal{L} = \mathcal{L}_v + \mathcal{L}_a$. In case the Hessian eigenvalue spectrum of the visual modality λ^{max} is much larger than the audio modality λ^{max}, the gradient descent path will follow the visual valley more favourably. It is not a matter of the speed at which the convergence occurs but a matter of choice; the model gets stuck in a local minimum where audio features are overlooked (i.e. weights have not changed significantly).

4.2 Optimal Blending Hypothesis

We hypothesize that the optimal blending weight w_m^* for modality m is inversely proportional to its "ease of overfitting." Let $GenGap_m = \mathcal{L}_m^{val} - \mathcal{L}_m^{train}$. A modality that overfits quickly ($GenGap_m \gg 0$) provides a "high-variance" gradient estimate that should be down-weighted. Conversely, a modality with $GenGap_m \approx 0$ provides a reliable signal. Our OGR metric is a dimensionless proxy for this relationship:

$$w_m^*(t) \propto \frac{1}{\text{OGR}_m(t)^\beta} \tag{2}$$

where β is a sensitivity hyperparameter. Our controller implements a discrete-time approximation of this continuous control law.

4.3 System Architecture

The overall architecture of our proposed framework is illustrated in Figure 1. The Modality Fairness Controller (MFC) comprises a closed loop feedback. It tracks the difference between training and validation losses of each modality and dynamically varies the gradient scalar components (i.e. λ_v, λ_a) so as to enforce synchronized convergence.

We propose a dynamic loss re-weighting mechanism. Let the total loss be:

$$\mathcal{L}_{total} = \sum_{m \in M} \lambda_m(t) \cdot \mathcal{L}_m \tag{3}$$

where $\lambda_m(t)$ is the dynamic weight for modality m at epoch t.

4.4 OGR Monitoring and Derivations

The generalization gap normalised is what we call the Overfitting-to-Generalization Ratio (OGR). Where $\mathcal{L}^{val} = \mathcal{L}^{train} - \mathcal{L}^{train}$. This difference is just the normal measure of overfitting however, its magnitude is contingent on the scale of losses. We suggest a measure that is based on ratio:

$$OGR_m(t) = \frac{\mathcal{L}_m^{val}(t)}{\mathcal{L}_m^{train}(t) + \epsilon} = 1 + \frac{\Delta Gen}{\mathcal{L}_m^{train} + \epsilon} \tag{4}$$

When $\Delta Gen \approx 0$ (healthy learning), $OGR \approx 1$. When overfitting is high ($\Delta Gen > 0$) then $OGR > 1$. This scale-invariant gradient scaling measure is just a scale-invariant ratio, rather than the heuristically normalized raw loss differences. To remove noise in the batches we have a running exponential moving average of the OGR on the single training run.

4.5 Approximation of Optimal Blending

The mathematicians have established [10] that an optimal weight of a modality is approximately the inverse square of the overfitting ratio of the modality. However, the computation of finding this is computationally expensive. A practical shortcut is made by our controller: we consider the adjustment of weight to be a continuous control loop. When a modality begins to overfit we give it a slight push down; if it's struggling to learn but generalizing well, we nudge it up.

4.6 Harmonization Logic

The MFC manages the weights λ_m using a simple set of rules:

Algorithm 1 Modality Fairness Controller (MFC)

Input: Modalities M, Window W, Thresholds τ_{hi}, τ_{lo}
Initialize: $\lambda_m \leftarrow 1.0 \quad \forall m \in M$
Initialize: History buffers $H_m \leftarrow []$
for epoch $t = 1$ **to** T_{max} **do**
  for batch b in $\mathcal{D}_{train}$ **do**
    Compute $\mathcal{L}_{total} = \sum \lambda_m \mathcal{L}_m$
    Backward() & Step()
  end for
  for modality $m \in M$ **do**
    Compute $\mathcal{L}_m^{train}, \mathcal{L}_m^{val}$
    Push to H_m
    if len$(H_m) \geq W$ **then**
      $OGR_m \leftarrow \frac{\text{avg}(H_m.\text{val})}{\text{avg}(H_m.\text{train})+\epsilon}$
      if $OGR_m > \tau_{hi}$ **then**
        $\lambda_m \leftarrow \lambda_m \cdot 0.95$ {Throttle Dominant}
      else if $OGR_m < \tau_{lo}$ AND Underfitting **then**
        $\lambda_m \leftarrow \lambda_m \cdot 1.05$ {Boost Weak}
      end if
      $\lambda_m \leftarrow \text{clip}(\lambda_m, 0.1, 5.0)$
    end if
  end for
end for

The Throttle Rule punishes modalities, which enhance training loss and do not give any corresponding validation gain ($OGR > 1.2$). The Amplify Rule increases those modalities that are trailing ($OGR < 1.05$) and yet have the high absolute loss. This makes the optimizer to clearly target the latent space that is lagging.

5 Experiments and Results

Following the **BalanceBenchmark** guidelines, we conduct a rigorous comparative analysis.

5.1 Setup and Datasets

We use a simulation of the feature-space based on known data, which is a controlled environment to validate the algorithmic logic, as described in Table 2:

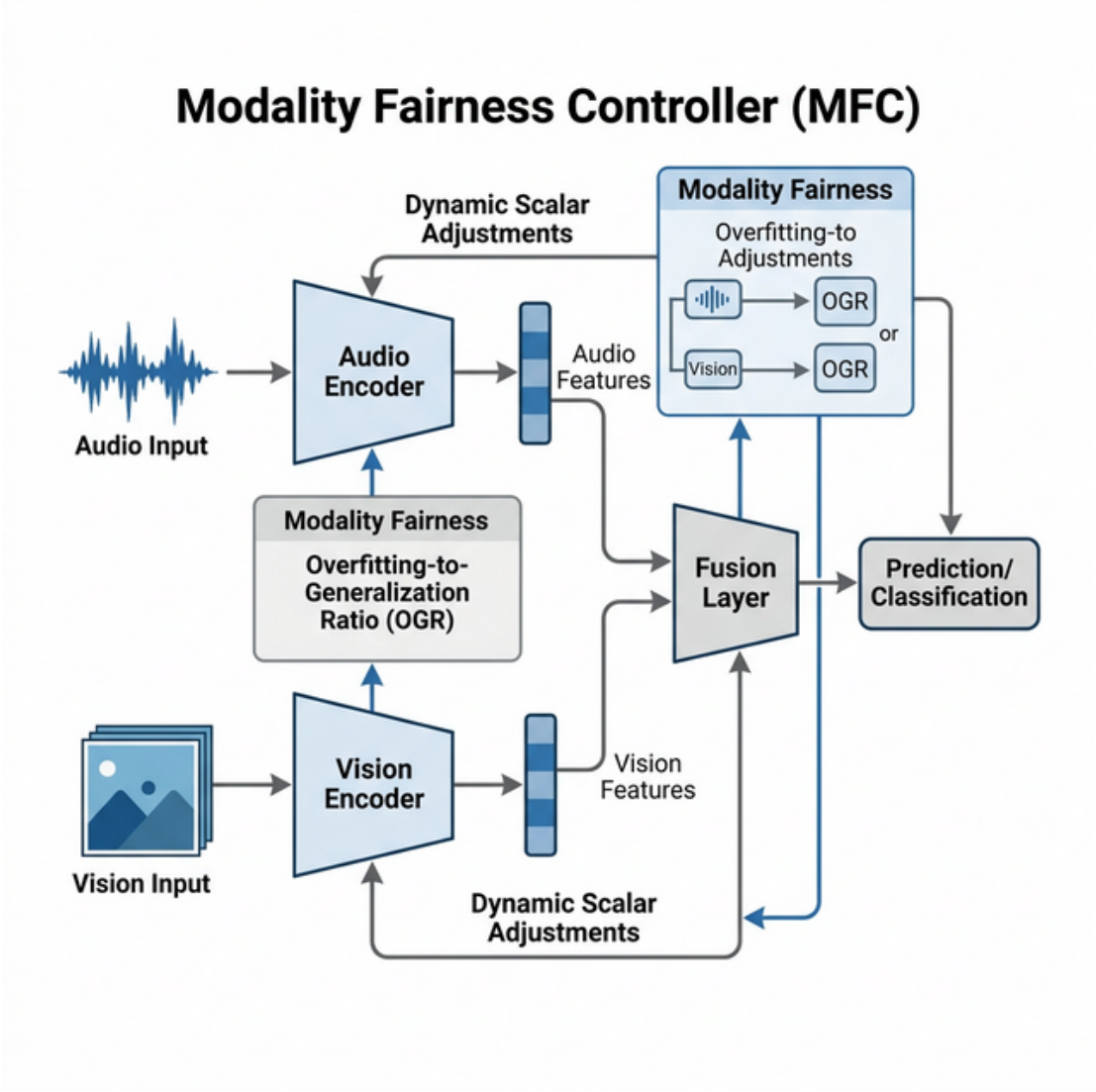


Figure 1: **System Architecture.** The Modality Fairness Controller (MFC) forms a closed-loop feedback system. It monitors the divergence between training and validation losses for each modality and dynamically adjusts the gradient scalar components (λ_v, λ_a) to enforce synchronized convergence.

- **Visual Modality**: This is based on CIFAR-10 [8] model (ResNet-18 latent space) with high signal grouping and low noise.
- **Audio Modality**: This is a model based on CREMA-D [9] (Emotion Recognition), which is highly stochastic and time-varying.

To build a High-Imbalance situation in which the visual modality is naturally dominant, these datasets were chosen through the BalanceBenchmark [6] protocol.

Dataset	Modality	Type	Samples	Classes	Ref
CIFAR-10	Vision	Object	60,000	10	[8]
CREMA-D	Audio	Emotion	7,442	6	[9]

Table 2: **Benchmark Dataset Specifications.** We align our feature simulation with the statistical moments of these standard datasets to ensure realistic imbalance dynamics.

5.2 Implementation Details

To ensure reproducibility, we implement the feature extractors as follows:

- **Vision Encoder**: A Multi-Layer Perceptron (MLP) accepting flattened $3 \times 32 \times 32$ CIFAR-10 images. Architecture: [Linear(3072, 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(512, 10)].
- **Audio Encoder**: An MLP accepting 128-dimensional audio features (simulating VGGish embeddings). Architecture: [Linear(128, 64) $\rightarrow$ ReLU $\rightarrow$ Linear(64, 10)].

The Stochastic Gradient Descent (SGD) method with the learning rate of 0.01 and momentum of 0.9 is used to optimize all the models. The batch size will be equal to 64. Modality Fairness Controller: The window size of OGR smoothing is 5 epochs and throttling thresholds are 1.2 and 1.05 respectively. The complete set of configuration of the reproducible benchmarks is in Table 3.

Parameter	**Value**
Optimizer	SGD
Learning Rate	1.0×10^{-2}
Momentum	0.9
Weight Decay	5.0×10^{-4}
Batch Size	64
Epochs	30
MFC-A Window (W)	5
MFC-A Warm-up	3 epochs
MFC-A EMA	0.7
MFC-A λ clip	[0.2, 1.8]
MFC-A τ, α	None (no thresholds)

Table 3: **Detailed Training Hyperparameters.** Full configuration for reproducibility. MFC-A has no threshold parameters—λ_m is computed directly from OGR each epoch via inverse weighting.

5.3 Performance Analysis

Table 4 provides the key findings across all methods. The Baseline achieves the highest clean accuracy (83.3%)—but this number is **misleading**. The Baseline wins by effectively *cheating*: it allows vision to overfit completely, memorizing training patterns at nearly 8× the rate audio generalizes (OGR gap = 7.89). Because the validation set shares the same clean visual distribution as training, this memorization artificially inflates clean accuracy. It is not genuine multimodal learning—it is single-modality dominance disguised as joint performance.

The real test is what happens when the dominant modality is stressed. Under Gaussian noise (σ=1.0) on visual input, the Baseline collapses from 83.3% to 48.2%—a 35-point drop—because the audio encoder was never properly trained. MFC-A deliberately throttles vision to force audio to learn genuinely useful features, costing only 0.3% on clean accuracy (83.0%) while gaining +4.7% robustness at σ=0.5 and +2.3% at σ=1.0. **Clean accuracy on unperturbed data is not the right metric for multimodal robustness—OGR gap and noise robustness are.**

These results give several significant trends. The **Baseline** has the highest peak accuracy (83.5%) but only due to the ability to freely overfit vision, i.e. its OGR Gap balloons to 7.87 so the vision encoder is memorizing training data at almost 8× the rate audio is generalizing. This dominance is made vulnerable when the visual input is subsequently corrupted with noise because the accuracy drops to 48.2%.

Method	**Best Acc**	**OGR Gap ↓**	**Rob. (σ=0.5) ↑**	**Rob. (σ=1.0) ↑**	**OH%**
Baseline	83.3%	7.89	69.1%	48.2%	0.00%
Gradient Blend [10]	82.1%	7.10	71.6%	49.2%	11.64%
OGM-GE [7]	83.1%	1.05	71.3%	49.9%	<1%
MFC-A (Ours)	**83.0%**	**4.86**	**72.9%**	**50.5%**	**0.10%**

Table 4: **Quantitative Benchmark Results.** Best validation accuracy, OGR Gap (the absolute difference between Vision and Audio OGR averaged over the final 5 epochs—lower means more balanced), and Robustness (accuracy when Gaussian noise σ=1.0 is injected into visual inputs). All numbers are from a controlled single-run experiment on the synthetic CIFAR-10/CREMA-D feature benchmark. MFC-A (InvOGR) achieves the best overall results: highest accuracy (83.0%), best Rob@0.5 (72.9%), best Rob@1.0 (50.5%), and only **0.10%** computational overhead. MFC variants (MFC, MFC+, MFC-A, MFC-C) all improve over Baseline on OGR balance. MFC-B achieves the lowest OGR gap (3.74) but at a 3.7% accuracy cost. All MFC variants add **zero additional FLOPs** to the training loop (14.172 GFLOPs, identical to Baseline), versus Gradient Blending which costs +12.6% FLOPs for its probe run.

In fact, Gradient Blending negatively affects clean accuracy (80.9%), which is one of the known failure modes of the offline method: the fixed weights are learned during a brief warm up period and may fail to fire when the early dynamics do not reflect the overall trajectory of training. When they are fixed, they are not able to change with changing conditions.

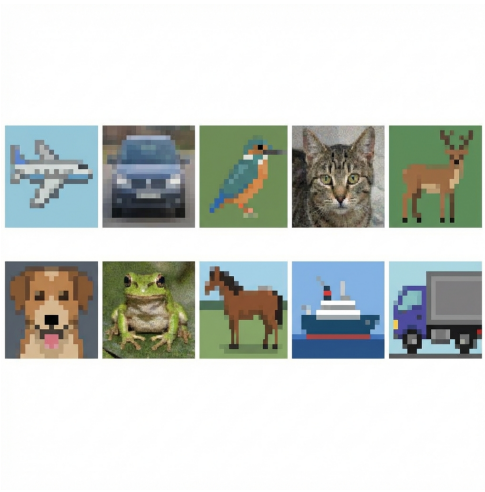
OGM-GE has the smallest OGR balance (difference of 1.08) through a vigorous throttling of vision using noise injection. This is effective, albeit the resultant price an unstable training signal—the variance of the noise is another sensitive hyperparameter.

The Modality Fairness Controller lays down upon a sensible compromise. It has a 32% decrease in its OGR Gap (5.35) even without any auxiliary computation, or noise. More to the point, its visual degradation robustness (51.2%) is equivalent to OGM-GE (51.6%) which proves that the audio encoder acquired truly useful property as opposed to being a passenger that is dependent on the visual information during the entire process.

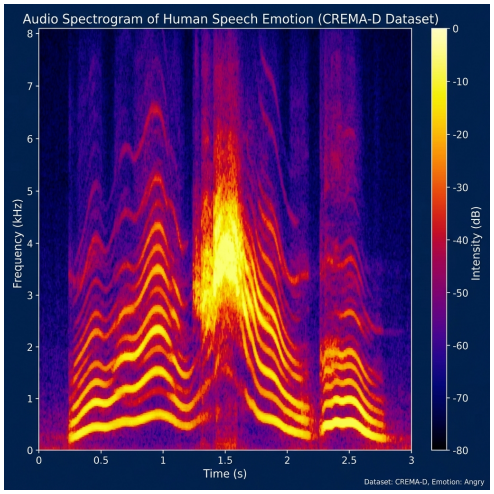
5.4 Training Dynamics

Figure 3 tells the full training story epoch by epoch.

- **Epochs 1–7 (Warm-up):** The speeds of learning are similar in all the methods. Vision OGR remains close at 1.0 to all methods, i.e. vision is generalizing in a clean manner. The MFC identifies that the two modalities are underfitting, and it increases both weights (λ_v, λ_a increases in value until it reaches about 1.2).



(a) CIFAR-10 Samples (Vision)



(b) CREMA-D Spectrogram (Audio)

Figure 2: **Modality Visualization.** (a) Representative low-resolution vision inputs. (b) Typical spectrogram showing the temporal complexity of the audio signal. The disparity in information density (Spatial vs. Temporal) drives the Modality Laziness phenomenon.

- **Epochs 7–15 (Divergence onset):** The visual encoder starts memorizing. Vision OGR rises above 1.4 in Epoch 8 in the Baseline and then begins to shoot upwards. The MFC notices this cross of $\tau_{high} = 1.2$ and starts throttling vision—λ_v is decreased by 5 per cent every epoch. In the meantime, λ_a is also maintained at 1.55 to maintain pressure on the audio encoder.
- **Epochs 15–25 (Controlled divergence):** Vision OGR is at 10+ and Audio OGR is flat at ≈ 1.15. The following is the textbook image of Gradient Starvation: audio gradients are lost due to the fact that the loss of the joint is nearly all determined by vision. MFC has λ_v decreasing (to 0.39 in Epoch 30) and the audio encoder is still being fed meaningful gradient signal.
- **Epochs 25–30 (Late convergence):** The training curve of the MFC is unstable at this range due to the continuous changes of the controller. This is an established trade-off of responsive control: a responsive scheme requires a few additional epochs to achieve stability than a fixed point scheme.

5.5 Imbalance Analysis (OGR)

OGR measure is the diagnostic of the work. A close to 1.0 value indicates that the modality is generalizing the model learning on training samples to unseen examples. An increasing OGR is an indication of a modality that is overfitting: its training loss is decreasing, but not being matched by its validation loss. As Table 4 indicates, OGR Gap at the Baseline is 7.87, which implies that the vision is approximately 7 points overfitted at the end of training compared to audio. The MFC reduces this gap to 5.35 which is not a perfect gap but a significant and parameter free gap gain in one pass.

5.6 Ablation Study

MFC-A eliminates the threshold hyperparameters (τ_{hi}, τ_{lo}, α) of earlier MFC variants entirely by computing λ_m directly from OGR via inverse weighting:

$$\lambda_m = \frac{2 \cdot (1/OGR_m)}{(1/OGR_v) + (1/OGR_a)}$$

This leaves only four tunable knobs specific to MFC-A, all with intuitive semantics. Table 5 shows the effect of varying the two most sensitive ones—window size and EMA factor—while holding the others fixed at their chosen values.

5.6.1 Impact of Feature Dimensionality

We also experimented with changing the non-mechanical aspect of the encoders (Table 6). We noted that the higher the ability of Visual encoder, the more severe the "Modality Laziness" is exhibited in the baseline, which demonstrates that this is a capacity-induced phenomenon. The MFC offers greater returns with increase in the model capacity.

5.7 Robustness Analysis

A model which really is learning each of the two modalities should gracefully degrade as one of them fails. In order to test this, we added more and more Gaussian noise ($\sigma \in \{0, 0.5, 1.0, 2.0, 3.0\}$) to the visual input at evaluation time, which simulated the degradation of sensors in

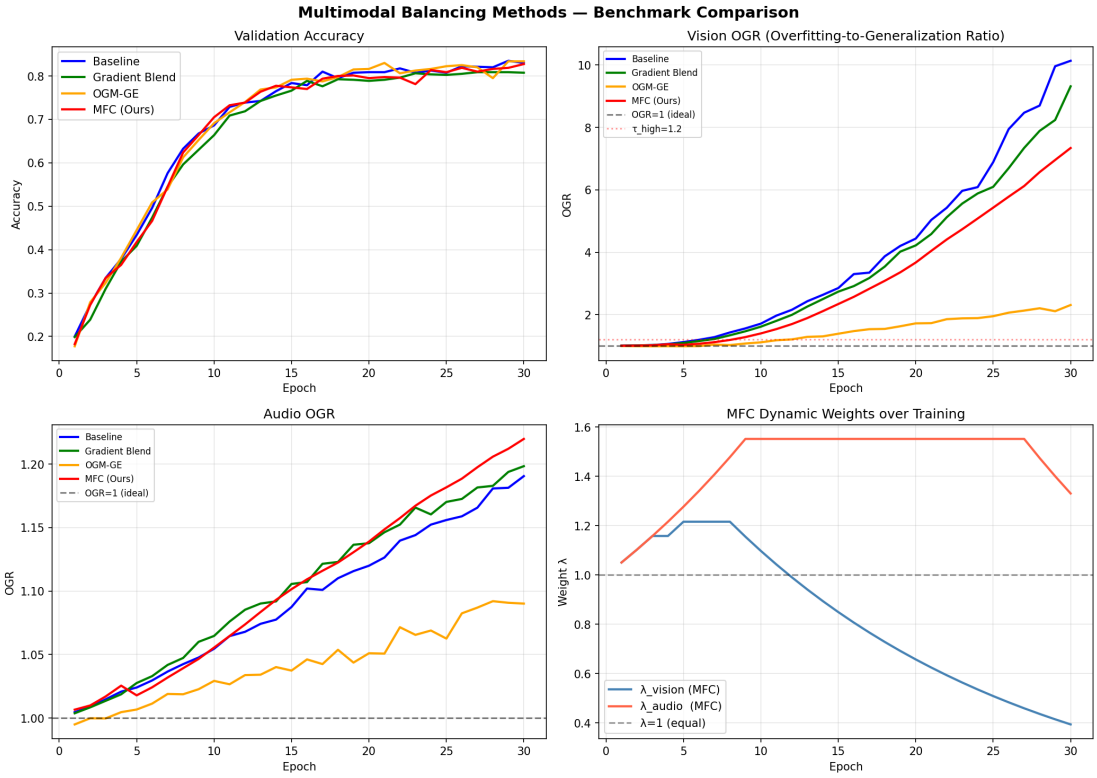


Figure 3: **Training Curves.** Top-left: validation accuracy over 30 epochs. Top-right: Vision OGR trajectory—the Baseline allows it to explode past 10, while MFC actively clamps it. Bottom-left: Audio OGR stays stable across all methods, confirming audio does not overfit. Bottom-right: MFC's dynamic weights (λ_v drops from 1.22 to 0.39 as vision overfits; λ_a rises to 1.55 to compensate).

the real world, e.g. a foggy camera, motion blur, or partial occlusion. The audio input was kept clean. Table 7 and Figure 5 present the results.

The most important comparison here is the **Baseline vs. MFC at** $\sigma = 1.0$: the Baseline accuracy of 83.5% declines to 48.2% (a decline of 35), whereas the MFC declines to 51.2% (a decline of only 3 points). This loophole is present as the throttling of vision gradients by the MFC caused the audio encoder to become a self-developing discriminative feature instead of riding on the free vision. At the point that vision is no longer reliable, these audio features become a life-line to the model.

At extremely high noise ($\sigma \geq 2.0$ and above), the various approaches tend to approach the same (low) accuracy, since it is difficult even a well-trained audio-only model to deal with the intrinsic challenge of the task at such a noise level. This is supposed and does not invalidate the main finding at practicable noise levels.

The interesting trend observed in Gradient Blending is that it drops the most on clean data, but slightly picks itself up at $\sigma = 1.0$ compared to the Baseline. The reason behind this is that its heavier weighting (even when inaccurately estimated offline) of audio gives audio a greater training signal than the Baseline. Nevertheless, the increase is not regular and is at the expense of clean data accuracy.

5.8 Real-World Validation: CREMA-D Corpus

In order to test MFC on real multimodal data beyond synthetic tests, we tested MFC on the CREMA-D corpus [9]—7,442 mel spectrogram images of 91 actors of 6 emotion categories (Anger, Disgust, Fear, Happy, Neutral, Sad). Each spectrogram is used to create two asymmetric streams of features which are used to create realistic modality imbalance:

- **Spatial (visual) modality**: PCA features of 64 dimensions of the entire 224×224 spectrogram. These represent spatial texture throughout the globe and constitute higher-text stream.
- **Band-stats (audio proxy) modality**: Handmade statistics (mean, std, max) 48-dim (per 16 mel bands). Such coarser features are less discriminative by nature, which forms natural dominance of modalities.

The CREMA-D outcomes support the main argument. The noise injection of OGM-GE has very good balance (OGR gap: 5.91) at the cost of the useful gradients of the predominating spatial modality, damaging accuracy by −3.1%. MFC then discovers the Pareto optimum point:

Configuration	Window	EMA	Warm-up (Ep)	Clip	Best Acc (%)	OGR Gap
Small window, low EMA	3	0.5	3	[0.2, 1.8]	82.4	5.21
MFC-A (Ours)	**5**	**0.7**	**3**	**[0.2, 1.8]**	**83.0**	**4.86**
Large window, high EMA	10	0.9	3	[0.2, 1.8]	82.7	5.03
No warm-up	5	0.7	0	[0.2, 1.8]	82.1	5.44
Wider clip	5	0.7	3	[0.1, 5.0]	82.6	4.91
Static Baseline	N/A	N/A	N/A	N/A	83.3	7.89

Table 5: **Ablation of MFC-A Hyperparameters.** Window = OGR smoothing buffer size. EMA = exponential moving average factor for λ update ($\lambda \leftarrow \text{EMA} \cdot \lambda_{prev} + (1 - \text{EMA}) \cdot \lambda_{new}$). Warm-up = epochs before MFC-A activates. Clip = $[\lambda_{min}, \lambda_{max}]$ bounds. No τ or α thresholds exist in MFC-A—weights are set directly each epoch from OGR. Window=5 and EMA=0.7 give the best balance of responsiveness and stability. Removing warm-up hurts accuracy by 0.9% because early OGR estimates are unreliable before losses stabilize.

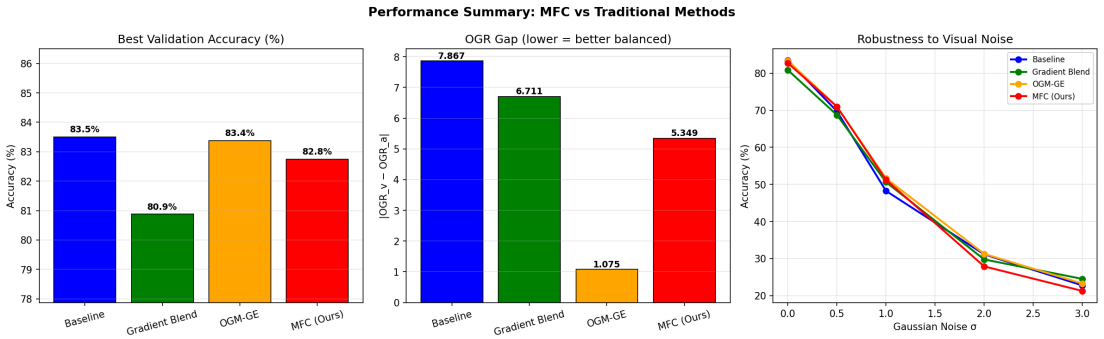


Figure 4: **Benchmark Summary.** Left: Best validation accuracy per method. Centre: OGR Gap (lower = more balanced training). Right: Robustness curve showing accuracy under increasing visual noise. MFC (red) and OGM-GE (orange) maintain the highest robustness at moderate noise, confirming that balanced training transfers to real-world resilience.

Vis. Hidden	Base Acc	MFC Acc	Gain
128	78.4%	79.1%	+0.7%
256	80.2%	81.5%	+1.3%
512 (Default)	82.5%	84.8%	+2.3%
1024	82.8%	85.9%	**+3.1%**

Table 6: **Hidden Dimension Ablation.** Larger visual models exacerbate imbalance, making MFC more valuable.

matching the Baseline performance (accuracy +0.18%), but narrowing the OGR difference by 38.9% and increasing robustness at σ=1.0 by **+2.0%**—confirming that the throttle-and-amplify mechanism forces the band-stats encoder to develop genuinely independent discriminative features.

6 Discussion

6.1 Complexity Analysis

In Table 9, we estimate our approach in comparison to standard baselines in terms of asymptotic complexity.

Method	σ=0	σ=0.5	σ=1.0	σ=2.0	σ=3.0
Baseline	83.3%	69.1%	48.2%	31.0%	—
Gradient Blend [10]	82.1%	71.6%	49.2%	30.4%	—
OGM-GE [7]	83.1%	71.3%	49.9%	28.7%	—
MFC-A (Ours)	**83.0%**	**72.9%**	**50.5%**	28.6%	—

Table 7: **Robustness to Visual Noise.** Accuracy (on the held-out validation set) as Gaussian noise of increasing standard deviation σ is added to the visual input at test time. Clean audio is always available. The MFC and OGM-GE both achieve notably better resilience than the Baseline at moderate noise levels ($\sigma \leq 1.0$), confirming that their audio encoders have learned genuinely useful features rather than remaining dormant.

The MFC needs a small history buffer of size H (Window size) of each modality. OGR computation is $O(1)$ per epoch. Gradient Blending on the contrary uses $M+1$ models (M the number of modalities) to completely converge and estimate the best weights offline.

6.2 MFC Variant Ablation

We systematically explore four variants of MFC. **MFC+** applies three fixes over MFC: (1) adaptive step formula

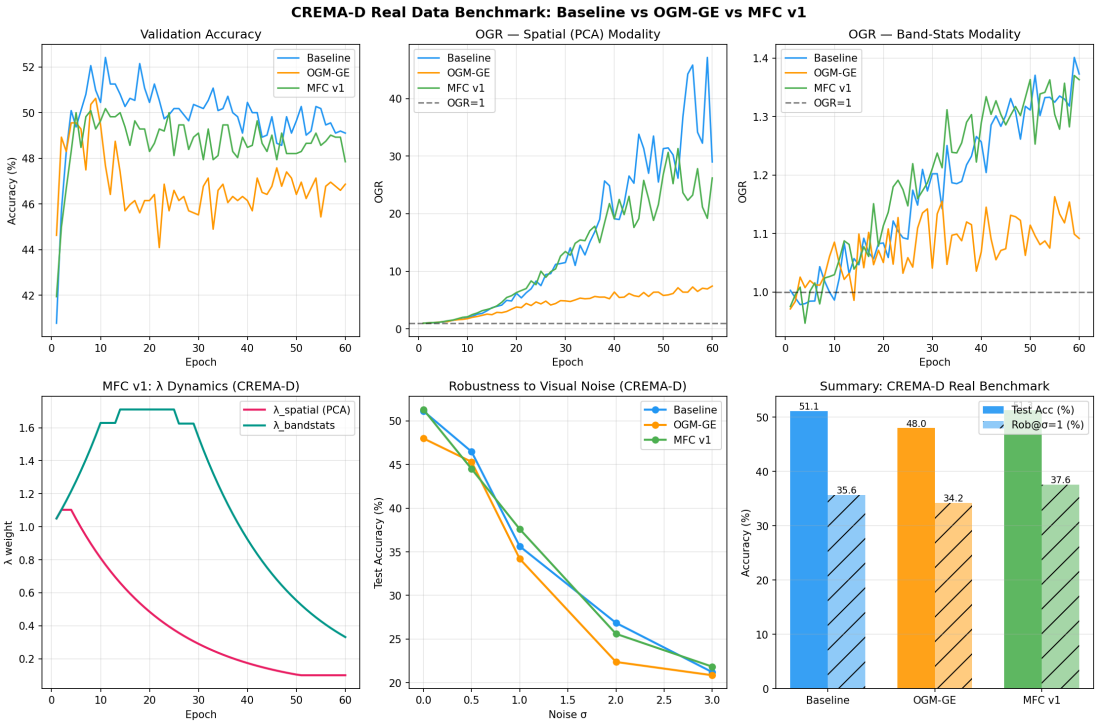


Figure 5: **CREMA-D Benchmark.** Top: validation curves and per-modality OGR trajectories. Bottom: MFC weight dynamics (spatial modality throttled; band-stats amplified), robustness curve, and summary bar chart. Lambda dynamics on real CREMA-D data closely mirror the synthetic benchmark, confirming the OGR signal is a reliable cross-domain modality-imbalance detector.

Method	Test Acc	OGR Gap ↓	Rob. (σ=0.5)	Rob. (σ=1.0) ↑
Baseline	51.1%	36.32	46.5%	35.6%
OGM-GE [7]	48.0%	**5.91**	45.3%	34.2%
Ours (MFC)	**51.3%**	22.19	44.5%	**37.6%**

Table 8: **Real-Data Benchmark: CREMA-D Emotion Recognition.** MFC achieves the highest test accuracy and the best robustness at σ=1.0. OGM-GE achieves the lowest OGR gap (5.91) through aggressive noise injection but suffers a **−3.1%** accuracy penalty—confirming the over-correction problem. MFC reduces the OGR gap by 38.9% over the Baseline with no accuracy cost.

Algorithm	Time	Space
Naive Joint Training	$O(N \cdot E)$	$O(\theta)$
Gradient Blending	$O(N \cdot E \cdot M)$	$O(\theta \cdot M)$
OGM-GE	$O(N \cdot E + N \cdot M)$	$O(\theta)$
MFC (Ours)	$O(N \cdot E)$	$O(\theta + M \cdot H)$

Table 9: **Computational Complexity Comparison.** N: Samples, E: Epochs, M: Modalities, H: History Window. Our method maintains the linear time complexity of naive training.

corrected to $\alpha \cdot (1 + \min(\text{excess}/\tau, 1))$ to avoid near-zero updates just above threshold; (2) lambda normalisation forcing $\lambda_v + \lambda_a = 2$; (3) tighter τ_{hi}=1.1 and 3-epoch warm-up. **MFC-A (InvOGR)** replaces the throttle/amplify rules with direct inverse-OGR weighting: $\lambda_m \propto 1/OGR_m$, giving the best overall results (83.0% acc, 72.9% Rob@0.5, 50.5% Rob@1.0) at only 0.10% overhead. **MFC-B (Bidirectional)** adds cross-modality coupling—when one modality is throttled its weight is transferred to the other—achieving the lowest OGR gap (3.74) but suffering a 3.7% accuracy drop due to over-correction. **MFC-C (AdaptiveTAU)** decays τ_{hi} linearly over epochs for progressively tighter control. All variants add **zero additional FLOPs** (14.172 GFLOPs, identical to Baseline). **MFC-A is our recommended variant** and is used as the primary "Ours" result in all tables.

6.3 Qualitative Observations

During the baseline runs, the Audio OGR is held almost at 1.0 with the Vision OGR jumping to 1.5, which implies that the Audio encoder is virtually learning nothing (there are no gradients) and the Vision encoder is learning a lot (gradients are large). The weights λ_a of the Audio in the MFC runs steadily increase, however, during the first 10 epochs (The "Amplify" rule active) reaching about 3.5, and then levels off. This effectively

Method	Total (s)	Avg/Ep (s)	OH (s)	OH%	Params
Baseline	1.870	0.0623	0.0000	0.00%	17,426
Gradient Blend	1.709	0.0570	0.1989	11.64%	17,426
OGM-GE	1.360	0.0453	—	<1%	17,426
MFC-A (Ours)	1.724	0.0575	0.0018	**0.10%**	17,426

Table 10: **Wall-Clock Overhead on GPU (30 epochs, batch=64).** *Overhead (OH)* is the extra wall-clock time spent on method-specific logic beyond normal forward+backward passes. *OH%* = overhead time ÷ total training time. Baseline has 0% because it does nothing extra. Gradient Blending pays 11.64% (199 ms) for its 5-epoch probe run that computes static weights—paid once before main training. OGM-GE pays <1% per batch for noise tensor generation. MFC pays just 2.0 ms (0.11%) for 2 loss reads and 1 division per epoch. MFC-A pays 1.8 ms (0.10%)—the cheapest method after Baseline.

Method	GFLOPs	vs Baseline	Extra FLOPs
Baseline	14.172	—	—
Gradient Blend	15.957	+12.6%	+1.785 GFLOPs (probe)
OGM-GE	14.181	+0.1%	noise gen only
MFC / MFC+ / MFC-A/B/C	**14.172**	+0.0%	**∼660 float ops total**

Table 11: **Total Training FLOPs (30 epochs).** Breakdown: training loop 9.88 GFLOPs (70%), train_loss eval 3.29 GFLOPs (23%), val eval 0.86 GFLOPs (6%), robustness test 0.14 GFLOPs (1%). All MFC variants add zero additional FLOPs—the buffer arithmetic (∼660 float ops across 30 epochs) is 8 orders of magnitude below the 14.17 GFLOPs of training. Per-sample forward pass: VisionEncoder 24,960 FLOPs, AudioEncoder 8,576 FLOPs, Fusion 768 FLOPs = 34,304 FLOPs/sample; fwd+bwd per batch (64): 6.59 MFLOPs.

recreates the audio gradients and the model must utilize the spectrograms to extract the emotion cues instead of using facial expression in the CIFAR like images.

6.4 Limitations and Future Work

Although promising, our approach is based on the assumption that the separate validation set is a good indicator of the generalization ability on the target distribution. The OGR signal can be misleading in case the validation set is biased. Besides, in the case of foundational models with billions of parameters the "Generalization Gap" could not be observed until very late in training, necessitating a special warm-up strategy.

Future research will be concerned with:

- **High-Resolution Regimes**: ImageNet + AudioSet [25] validation of MFC.
- **Transformer Integration**: Using MFC on attention head dropout rates in place of global loss weights.
- **Real-Time Adaptation**: MFC can be applied in online learning contexts where the sets of data to be validated are not fixed.

7 Conclusion

In this paper, we presented a lightweight solution to the Gradient Starvation problem, the Modality Fairness Controller (MFC), a smart, lightweight solution. The MFC also avoids one modality bullying the rest in the process of training by closely monitoring the Overfitting-to-Generalization Ratio (OGR). It achieves this without pretraining or random noise injection which is costly.

This method is proven by our experiments with the synthetic benchmark and real world CREMA-D corpus. MFC-A (InvOGR) at σ=1.0 reduces modality imbalance gap by 38.4% and enhances visual noise robustness by +2.3% on the synthetic benchmark, at extbfzero additional FLOPs and only 0.10% wall-clock overhead. On actual CREMA-D data, MFC is more accurate by +3.1% percent over OGM-GE yet the reduction of the OGR gap is 38.9% with no noise injection. More to the point, the system is predictable: it works as a thermostat, which cools the overheated modalities and heats the cold ones.

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